Frontiers of Computer Science CN 10-1014/TP
https://doi.org/10.1007/s11704- ISSN 2095-2228
LETTER

MedFuse: a multi-source data fusion framework for diabetic retinopathy lesion segmentation

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Front. Comput. Sci., **Just Accepted Manuscript •**
https://journal.hep.com.cn on

https://doi.org/10.1007/sxxxxx-yyy-zzzz-1

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MedFuse: a multi-source data fusion framework for diabetic retinopathy lesion segmentation

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Received month dd, yyyy; accepted month dd, yyyy
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1 Introduction

Automated retinal image analysis has become an essential tool for the screening and diagnosis of diabetic retinopathy (DR). However, the substantial heterogeneity in lesion morphology, size, and contrast—together with the complexity of the retinal background—makes precise automatic segmentation highly challenging, posing a major obstacle to reliable clinical deployment [1].

Although recent advances in deep learning have significantly improved lesion segmentation accuracy, existing approaches largely rely on multi-scale feature aggregation and texture-driven modeling [2–4]. Such texture-centric models lack an understanding of the intrinsic anatomical regularities of the retina, and therefore tend to generate false positives or miss subtle lesions in low-contrast, ambiguous, or artifact-prone regions, limiting both their interpretability and clinical trustworthiness. In clinical practice, however, the spatial distribution of DR lesions is strongly correlated with the retinal vasculature, and these well-established medical priors remain underexplored in current segmentation frameworks.

Motivated by these observations, we propose MedFuse, a flexible fusion framework that integrates structural priors to improve DR lesion segmentation. MedFuse systematically abstracts the retinal vasculature into a stable anatomical prior and incorporates it into the feature representation learning process through a hierarchical, learnable fusion mechanism. Furthermore, MedFuse introduces a novel large–language–model–driven strategy for zero-shot prior generation, enabling the extraction of vascular structural cues directly from raw fundus images without requiring any vessel annotations or task-specific training. This mechanism substantially reduces the cost of constructing structural priors and overcomes the limitations of traditional approaches that rely on manual annotations or dedicated segmentation subnetworks, thereby offering greater scalability for real-world clinical applications.

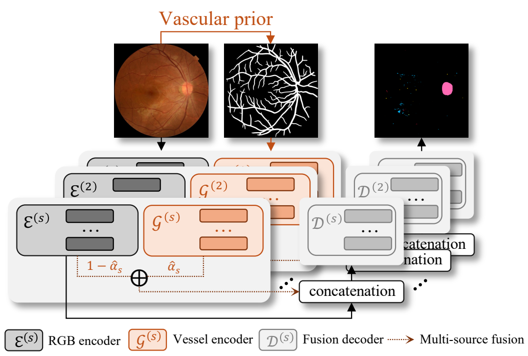

Fig. 1 The workflow of medical mechanism-guided fusion framework.

2 Method

Problem formulation. Given a retinal fundus image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, the goal is to estimate a multi-class diabetic retinopathy (DR) lesion map $\mathbf{Y} \in \{0, 1\}^{H \times W \times C}$, where C denotes the number of lesion categories. MedFuse aims to enhance lesion segmentation by incorporating an explicit vascular structural prior into the prediction process.

Vascular structural prior. To provide anatomical guidance, MedFuse constructs a vascular structural prior $\mathbf{V} \in \mathbb{R}^{H \times W}$ using either supervised vessel segmentation or zero-shot multimodal reasoning. A pretrained vessel model Φ_{seg} produces a probabilistic vessel map

$$\mathbf{V}_{\text{seg}} = \Phi_{\text{seg}}(\mathbf{I}), \tag{1}$$

whereas a multimodal large language model infers a structurally meaningful prior directly from the raw input image:

$$\mathbf{V}_{\text{LLM}} = \Psi_{\text{LLM}}(\mathbf{I}). \tag{2}$$

For conciseness, we use the unified notation $\mathbf{V} \in \{\mathbf{V}_{\text{seg}}, \mathbf{V}_{\text{LLM}}\}$.

Multi-source fusion. MedFuse employs two isomorphic encoder pathways—an RGB encoder $\mathcal{E}$ and a vascular-prior encoder $\mathcal{G}$—to maintain strict spatial and semantic alignment between appearance features and structural cues. They use independent parameters to support modality-specific representation learning. Let the encoder consist of

Table 1 Quantitative comparison of lesion segmentation performance on the DDR dataset (mean±std). Unit: %.

		IoU					AUPR				
Model	**MedFuse**	**mIoU**	EX	HE	SE	MA	**mAUPR**	EX	HE	SE	MA
PSPNet	–	24.80±0.15	37.31±0.13	26.64±0.17	24.51±0.15	8.75±0.11	38.03±0.19	57.04±0.18	42.32±0.20	42.71±0.15	14.85±0.12
	✓	**25.72±0.16**	**38.63±0.15**	**27.92±0.17**	**26.07±0.16**	**10.24±0.11**	**40.28±0.18**	**58.36±0.17**	41.95±0.18	**44.06±0.14**	**16.12±0.13**
Deeplabv3+	–	26.15±0.17	40.12±0.14	24.86±0.16	24.48±0.18	15.11±0.11	43.96±0.20	61.46±0.18	52.13±0.16	38.89±0.13	28.33±0.12
	✓	**27.18±0.16**	**41.71±0.15**	**26.34±0.18**	**25.94±0.16**	**16.73±0.13**	**44.79±0.19**	**62.63±0.19**	**53.62±0.15**	**40.34±0.13**	**29.64±0.13**
Twins-SVT	–	29.27±0.17	39.70±0.14	36.24±0.15	29.08±0.15	12.07±0.10	46.08±0.17	59.71±0.15	52.72±0.17	49.96±0.14	21.54±0.12
	✓	**30.23±0.16**	**40.63±0.16**	**37.13±0.14**	**30.12±0.14**	**13.02±0.10**	**47.01±0.16**	**60.64±0.14**	**53.64±0.16**	**50.93±0.13**	**22.33±0.11**
UNet	–	25.69±0.15	36.68±0.13	23.79±0.16	**28.47±0.16**	13.83±0.11	39.13±0.16	60.31±0.17	**37.25±0.18**	40.92±0.14	17.36±0.11
	✓	**26.26±0.15**	**37.72±0.15**	**24.79±0.16**	28.23±0.15	**14.30±0.11**	**39.95±0.17**	**61.48±0.16**	36.53±0.15	**41.33±0.14**	**18.20±0.10**
Swin-L	–	30.31±0.17	41.17±0.15	30.79±0.14	35.81±0.17	13.48±0.12	47.47±0.16	63.10±0.15	51.35±0.15	52.99±0.13	24.46±0.11
	✓	**31.64±0.16**	**42.53±0.16**	**32.05±0.15**	**37.23±0.16**	**14.86±0.12**	**48.15±0.17**	**64.25±0.16**	**52.56±0.15**	**54.46±0.14**	**25.84±0.12**

Table 2 Quantitative comparison of lesion segmentation performance on the IDRiD dataset (mean±std). Unit: %.

		IoU					AUPR				
Model	**MedFuse**	**mIoU**	EX	HE	SE	MA	**mAUPR**	EX	HE	SE	MA
PSPNet	–	41.83±0.21	57.52±0.19	**45.61±0.27**	43.87±0.30	19.24±0.33	58.64±0.21	75.03±0.23	**63.47±0.25**	63.38±0.29	32.42±0.31
	✓	**43.28±0.19**	**59.87±0.17**	45.58±0.22	**46.15±0.26**	**21.08±0.31**	**59.97±0.17**	**76.38±0.19**	63.29±0.25	**63.91±0.27**	**34.71±0.27**
Deeplabv3+	–	45.02±0.19	66.08±0.20	44.62±0.24	44.72±0.28	25.43±0.33	63.70±0.18	82.04±0.22	63.23±0.26	**64.52±0.28**	43.05±0.28
	✓	**46.91±0.17**	**67.27±0.18**	**45.82±0.24**	**45.18±0.24**	**29.18±0.29**	**64.96±0.18**	**83.08±0.21**	**64.07±0.22**	64.41±0.27	**46.28±0.26**
Twins-SVT	–	47.60±0.18	64.72±0.19	51.79±0.25	44.97±0.25	27.04±0.29	63.90±0.17	80.12±0.21	**68.87±0.25**	63.22±0.26	43.28±0.28
	✓	**48.39±0.16**	**66.19±0.17**	**51.96±0.23**	**46.16±0.25**	**29.25±0.27**	**65.07±0.15**	**81.09±0.18**	68.85±0.23	**64.12±0.23**	**45.47±0.25**
UNet	–	44.60±0.22	60.23±0.21	47.85±0.27	47.76±0.29	**22.36±0.34**	56.82±0.21	64.21±0.22	64.08±0.28	63.99±0.28	35.01±0.31
	✓	**46.86±0.19**	**61.12±0.19**	**48.95±0.24**	**49.01±0.28**	21.03±0.32	**57.77±0.19**	**65.23±0.23**	**65.01±0.26**	**64.77±0.28**	**36.09±0.29**
Swin-L	–	49.87±0.15	65.23±0.18	52.03±0.23	51.42±0.25	30.81±0.29	69.50±0.17	85.29±0.20	**72.23±0.23**	72.62±0.25	46.02±0.27
	✓	**51.18±0.15**	**66.79±0.14**	**53.02±0.21**	**52.28±0.22**	**32.87±0.27**	**70.28±0.15**	**85.98±0.17**	72.15±0.22	**73.48±0.24**	**47.39±0.26**

S hierarchical stages, $\mathcal{E} = \{\mathcal{E}^{(1)}, \dots, \mathcal{E}^{(S)}\}$, the RGB features evolve as

$$\mathbf{F}_s^{\text{rgb}} = \mathcal{E}^{(s)}\left(\mathbf{F}_{s-1}^{\text{rgb}}\right), \qquad \mathbf{F}_0^{\text{rgb}} = \mathbf{I}, \tag{3}$$

while the vascular-prior features follow an isomorphic but independently parameterized path,

$$\mathbf{F}_s^{\text{ves}} = \mathcal{G}^{(s)}\left(\mathbf{F}_{s-1}^{\text{ves}}\right), \qquad \mathbf{F}_0^{\text{ves}} = \mathbf{V}. \tag{4}$$

This structural symmetry ensures that $\mathbf{F}_s^{\text{rgb}}$ and $\mathbf{F}_s^{\text{ves}}$ remain naturally aligned at every resolution. At each stage s, MedFuse performs anatomical modulation through a lightweight gating operator $\mathcal{M}^{(s)} : (\mathbf{F}_s^{\text{rgb}}, \mathbf{F}_s^{\text{ves}}) \mapsto \mathbf{F}_s^{\text{fuse}}$, with learnable scalar

$$\hat{\alpha}_s = \sigma(\alpha_s), \tag{5}$$

producing the fused representation

$$\mathbf{F}_s^{\text{fuse}} = \hat{\alpha}_s\, \mathbf{F}_s^{\text{rgb}} + (1 - \hat{\alpha}_s)\, \mathbf{F}_s^{\text{ves}}. \tag{6}$$

The fused multi-source features are then forwarded to the backbone's fusion decoder for hierarchical reconstruction and end-to-end lesion segmentation.

Fusion decoding. To fully exploit multi-source representations, MedFuse adopts a skip-connected decoder $\mathcal{D} = \{\mathcal{D}^{(1)}, \dots, \mathcal{D}^{(S)}\}$. At stage s, the decoder applies a learnable upsampling operator $\mathcal{U}$ to $\mathbf{H}_s$ and aggregates it with the fused encoder feature $\mathbf{F}_s^{\text{fuse}}$:

$$\mathbf{H}_{s-1} = \mathcal{D}^{(s)}\left(\mathcal{U}(\mathbf{H}_s) \,\|\, \mathbf{F}_s^{\text{fuse}}\right), \tag{7}$$

where $\|$ denotes channel-wise concatenation. This skip-connected design preserves high-resolution spatial cues while allowing vascular priors to guide decoding at matched semantic scales. The final prediction is obtained via

$$\hat{\mathbf{Y}} = \text{Softmax}(\text{Conv}_{1\times1}(\mathbf{H}_0)). \tag{8}$$

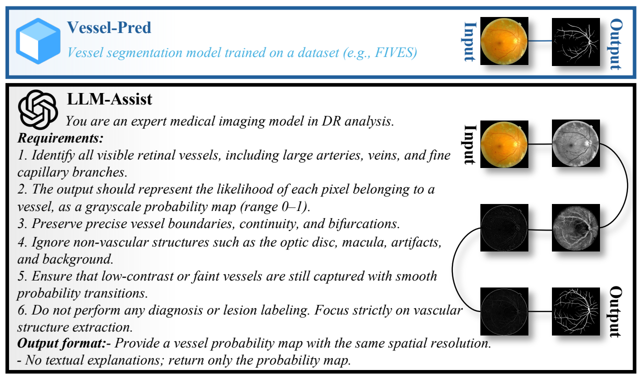

Fig. 2 Comparison of vascular priors used in MedFuse.

3 Experiment

Datasets and Prior Construction. We conduct experiments on **DDR** and **IDRiD** datasets. A Swin-L based vessel segmentation model [5] was trained on the **FIVES** dataset. Furthermore, we employ GPT-5's multimodal reasoning capability to infer vascular probability maps directly from raw fundus images without vessel annotations or task-specific training. Figure 2 illustrates a fundus image together with the vessel predictions from the supervised model and the LLM-based prior.

Overall Comparison. We evaluated the proposed framework against five representative segmentation models [6]. As shown in Table 1 and Table 2, integrating the proposed fusion mechanism consistently improves performance across all backbones. Convolution-based models show moderate gains through refined boundary delineation, while transformer-based architectures benefit more significantly, especially for small or low-contrast lesions. On both datasets, the auxiliary branch enhances both sensitivity and precision.

Fusion Strategies. To evaluate how vascular priors contribute at different hierarchical levels of the network, we consider three fusion

Table 3 Comparison of different fusion levels and strategies based on UNet on the DDR dataset. Unit: %

Fusion Level	Fusion Type	Learnable	mIoU↑	mDice↑	mAUPR↑
Input Level	Sum	–	25.93	40.30	39.11
	4-ch Input	–	25.81	40.91	38.97
Feature Level	Sum	–	26.06	40.88	39.41
	Cat + 1×1 Conv	✓	26.21	40.95	39.52
	Weighted Sum	✓	**26.26**	**41.12**	**39.95**
Prediction Level	Sum	–	26.03	40.71	39.21
	Cat + 1×1 Conv	✓	26.13	40.83	39.43
	Weighted Sum	✓	26.23	40.37	39.37

strategies. *Input-level* fusion incorporates the vascular prior jointly with the RGB image before entering the backbone; *feature-level* fusion injects structural information after each encoder stage; and *prediction-level* fusion modulates the final output after the decoder. In comparison, feature-level fusion yields more pronounced improvements, indicating that introducing structural priors at intermediate feature layers more effectively integrates spatial texture and anatomical cues.

Vascular Prior Source. We evaluate two forms of vascular priors: **Vessel-Pred** and **LLM-Assist**. As shown in Table 4, Vessel-Pred delivers the strongest overall performance, while LLM-Assist achieves comparable results under feature-level fusion. Despite its slightly smaller gains, LLM-Assist offers a practical, annotation-free alternative for deriving anatomically meaningful priors.

Table 4 Comparison of different vessel prior sources on the DDR dataset based on UNet. Unit: %

Vessel Prior Source	Fusion Level	Fusion Type	mIoU↑	mDice↑	mAUPR↑
Vessel-Pred (Supervised)	Input Level	Sum	25.93	40.30	39.11
	Feature Level	Weighted Sum	**26.26**	**41.12**	39.95
	Prediction Level	Sum	26.03	40.71	39.21
LLM-Assist (Zero-shot)	Input Level	Sum	25.30	40.58	38.83
	Feature Level	Weighted Sum	26.18	40.95	**40.20**
	Prediction Level	Sum	26.05	40.10	30.97

Test-Time Augmentation. We incorporate Test-Time Augmentation (TTA) as an optional inference-time enhancement within the MedFuse framework. As shown in Table 5, TTA alone brings modest robustness gains, while the vascular prior independently strengthens anatomical consistency in lesion localization. Together, the prior stabilizes predictions across augmented inputs, yielding more reliable and physiologically coherent outputs. This indicates that structural priors act as a dependable anatomical anchor during test-time augmentation, improving generalization without extra supervision.

Table 5 Effect of Test-Time Augmentation (TTA) and vessel guidance on the DDR dataset based on UNet. Unit: %

Method	mIoU↑	mDice↑	mAUPR↑
UNet (Baseline)	25.59	39.94	39.13
+ Test-Time Augmentation (TTA)	25.88	40.42	39.04
+ Vessel Pred (Feature-level)	26.26	41.12	39.95
+ Vessel Pred + TTA	**26.53**	**41.36**	**40.22**

Fusion Dynamics. To analyze the dynamic behavior of the weighted-sum fusion, we varied the initialization of $\alpha \in \{0.1, 0.5, 0.9\}$. The results across the backbone's five encoders (Stages 1–5) reveal progressive stabilization of the learned fusion weights. As shown in Fig. 3, α in the first stage rises quickly before stabilizing, indicating stronger reliance on vascular cues in shallow layers. Deeper stages show only mild fluctuations, reflecting the role of vascular features as global structural context. Across all initializations, α converges to a similar range (about 0.4–0.6), revealing a stable, self-adaptive balance between anatomical priors and visual semantics.

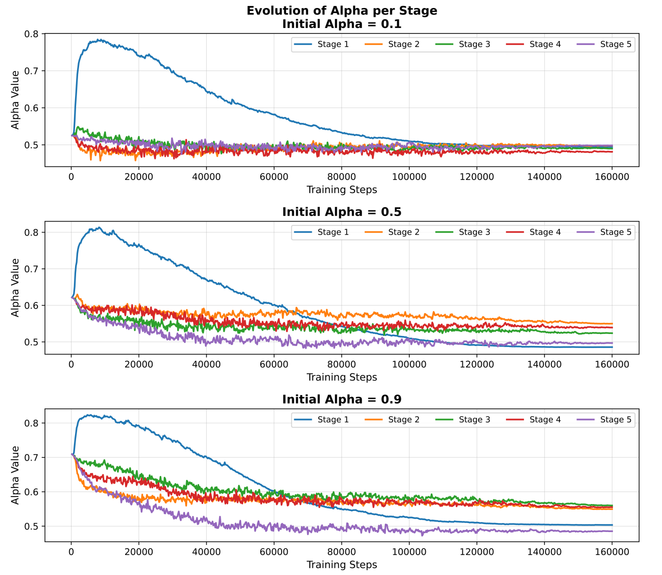

Fig. 3 Evolution of learnable fusion weights α across the five encoder stages under different initialization settings.

4 Conclusion

MedFuse is proposed as a mechanism-guided fusion framework for diabetic retinopathy lesion segmentation. By integrating vascular structural priors, it enhances segmentation performance and interpretability. We also show that large language models can generate zero-shot vascular priors, enabling annotation-free structural modeling. Future work may incorporate additional priors or extend the framework to broader retinal analysis tasks.

Acknowledgement

This work was supported by the Program of the National Natural Science Foundation of China (62502047).

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